

Quantum theory from sealed records I: Born composition, Lorentz signature, and the arrow of time

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Abstract

We present the foundational layer of a research program that reconstructs quantum-mechanical, relativistic, and thermodynamic structure from a single ontology: a ledger of *sealed records*, governed by three axioms — **R** (laws are laws of whole sealed histories), **S** (no distinction without a record), and **C** (couplings are fixed by self-consistency under refinement). The program builds on the stochastic–quantum correspondence of Barandes’ indivisible processes, replacing configuration trajectories by committed records as the primitive. Main results at stated scope, each with a machine-verified receipt in this submission or a tagged companion/corpus pointer ([C]/[P]): (i) the **Born layer** — record weights compose through square roots, and the tame-class reconstruction is *argued to* recover the Weyl algebra and to land in the Schrödinger representation via the Stone–von Neumann import, from record towers (at tame scope: the tame-class definition with its growth constants and the boundary classification are corpus-bound [P]), with ladder operators, selection rules, and exact continuum targets approached at machine precision; (ii) **Lorentz signature** — the commitment structure of records *is argued to select* a Lorentzian $(1, d)$ split with derived orientation classes (a structural argument from the commit-order asymmetry, its proof obligation corpus-bound [P]; the spatial dimension $d = 3$ is *selected*, by a minimality principle graded as input in the text, not derived), and the operational identification lapse = local clock rate is verified to 3.7×10^{-8} in curved record geometries (a model-internal consistency receipt), with first 3+1-dimensional cosmological checks (qualitative; no cosmological number is claimed in this batch); (iii) the **arrow of time** — reflection positivity is a *theorem* for eventless sectors with finite primitive Markov presentations, the entropy-production functional is the path-measure relative entropy $\sigma = D(P_{AB} \| P_{BA})$ (the classical identity of stochastic thermodynamics, here in record-pair form), and the presentation hypothesis is *sharp*: an explicit reversible, valid, finite-rank process (the “record clock”) fails reflection positivity, separating record laws from general reversible processes; (iv) **thermality** — horizon temperature $T = \kappa/2\pi$ is recovered in exact lattice constructions, presented in full in the companion [C-VIII] and cited here only as a consistency credential. We state precisely what is axiom, what is theorem, what is named identification, and what is imported mathematics — the full dependency audit is part of the paper — and we delimit the program’s frontier: the tame class, the regularity stratum of the emergent continuum, and continuous sector families.

1. Introduction

The starting point. Barandes’ stochastic–quantum theorem [1] shows that unitary quantum dynamics on a finite-dimensional Hilbert space corresponds exactly to *indivisible* stochastic processes — processes specified by transition data over whole time intervals rather than composable infinitesimal steps. This reopens a question usually closed by fiat: which classical-looking ontology, if any, stands behind quantum theory’s formal apparatus?

This program proposes an answer with a different primitive: not trajectories, not measurement outcomes, but **records** — definite, sealed entries in a ledger. Three axioms govern them:

- **Axiom R (records).** Physical law assigns weights to whole sealed histories of the ledger; “state at a time” is derived bookkeeping.
- **Axiom S (silent-seam exclusion).** Any datum that no sealed record registers — a silent label, phase, or holonomy — is not part of the physics; constructions requiring one are excluded or must be quotiented to gauge.
- **Axiom C (refinement self-consistency).** Records may be refined into finer records; the law’s couplings are fixed points of the requirement that refined ledgers obey the same law.

R and S are principle-type axioms in Einstein’s sense — they forbid rather than construct; C is of a different kind, a fixed-point *selection* condition, and we do not claim the principle-theory mantle for it. The program’s results are *forcing arguments* — some proved as theorems, others structural arguments resting on named inputs or corpus-bound proof obligations, as the Section 7 audit records: structures usually postulated (Hilbert space, signature, reflection positivity, fermions, gauge groups) appear as the survivors of the exclusions, *under the inputs and scope premises each layer names*. This first paper presents the kinematic, relativistic, and thermodynamic layer; a companion (Part II) treats statistics and gauge structure; further papers treat the matter-content selection and its falsifiable consequences.

Mathematical home, stated first. So that nothing rests on undefined vocabulary, the program’s primitives have concrete mathematical referents. Two citation registers are used, and we keep them honestly separate: [C0]–[C4], [C-VIII], [VII], and [IX] are companion papers *submitted together with this one* ([C0] the binding-theory companion; [C1]–[C4] necklace invariants, reflection positivity and realization, the moment test, the graded Weyl law; [C-VIII] horizons; [VII] the neutrino registration note; [IX] the mechanism paper), and [P] marks results whose full treatment lives in the program corpus — a longer, internally dated paper series available as supplementary material but *not* part of this submission. Every claim below whose proof is corpus-bound is tagged [P], never [C]; the [C] pointers are checkable companions and the [P] pointers are stated dependencies, not verified ones. The referents: a **ledger** is a stationary stochastic process of committed symbols (“records”) with its word-probability functional; **reflection structure** means bond- and site-reflection positivity of that functional (for the reconstruction tradition see [3, 7]), *jointly* equivalent ([C2], Thm. 2.1) to a representation by positive-semidefinite letter operators — site-RP is exactly what makes the letters positive operators, and the two hypotheses are not interchangeable (bond-RP alone does not give the operator form) — the operator form in which all of this paper’s transfer-matrix statements live; **seams** are the invariant tensors coupling letter fibers; record **towers** are inductive families of such processes with stated growth conditions (the “tame class,” defined where used). Sections below give,

for each result, the precise mathematical statement at this level, the receipt, and the pointer to the companion proof; this paper’s own role is the physical layer — which structures are *forced* and which are chosen.

Method. Every quantitative claim carries a machine-checked numerical receipt: a deterministic, fixed-seed script whose reruns are bit-identical and whose output is quoted directly. (We do not claim proof-assistant formalization; “receipt” means a reproducible computation, and theorems are proved as theorems in the text or companions.) Every load-bearing assumption is named and typed: *axiom*, *theorem*, *identification* (a bridge chosen, not proved), or *import* (classical mathematics). The audit appears in Section 7 — including the honest count: three axioms, a handful of named identifications, four scope premises, and three named structural inputs (the power-sum weight form; screen-transport linearity and transitivity; the spatial $d = 3$ minimality selection).

2. The Born layer at tame scope

What is and is not claimed. We do not offer a Gleason-style or decision-theoretic derivation of the Born rule, and we position this section explicitly against that literature [8–11]: probabilities here are *primitive* (record weights), as in Barandes’ correspondence [1], and the derived content is structural — that the weight calculus of sealed histories organizes itself into the Hilbert-space formalism, with amplitudes as square roots of weights and the operator algebra emerging from record towers. The quantum-reconstruction programs [10, 11] derive the formalism from operational axioms about laboratories; this program derives it from an ontology with no observers in it. Where they meet — the uniqueness of the quadratic weight calculus — we rely on the correspondence theorems, not on novelty claims.

What the “quadratic” imports, exactly. Although we do not *derive* the Born exponent, we can say precisely which single structural assumption it is. Let a record screen carry a finite complex weight vector $a = (a_1, \dots, a_n)$ and let the event weight be componentwise, $W_p(a) = \sum_i |a_i|^p$; require it invariant under the admissible screen transports. This separates into two layers. *Layer 1 (tautological)*. Once the screen norm is *declared* to be ℓ_q , the readable exponent is pinned to $p = q$ with no further content: an equal-split transport $(1, 0, \dots, 0) \mapsto n^{-1/q}(1, \dots, 1)$ sends the unit input ($W_p = 1$) to $W_p = n^{1-p/q}$, and a single screen size $n \geq 2$ already forces $n^{1-p/q} = 1$, i.e. $p = q$. This holds for *every* q and says nothing about *which* q — the readable exponent is *identically* the norm one declares. *Layer 2 (the substantive step)*. All the content is in *which* transports are admissible: the equal-split map preserves the ℓ_q norm of its one input–output pair, but for $q \neq 2$ it does **not** extend to a linear ℓ_q isometry of the screen at all, so its very admissibility as a transport is already a $q = 2$ statement. And there is a classical reason the admissible (linear, transitive) transports are the $q = 2$ (unitary) family: by the **Banach–Lamperti theorem** [17], for $q \neq 2$ the surjective linear isometries of complex ℓ_q^n are the generalized permutations (coordinate permutations dressed with phases), which do **not** act transitively on the screen sphere — none carries $(1, 0, \dots, 0)$ to the uniform superposition $n^{-1/q}(1, \dots, 1)$ — whereas $q = 2$ is the unique exponent whose isometry group $U(n)$ acts **transitively**. So the content of “the weight calculus is quadratic” reduces to three structural inputs: **(0)** the screen weight is a componentwise power-sum $W_p(a) = \sum_i |a_i|^p$ (additive over components, phase-blind per component, a single global exponent — itself an assumed form, not derived from R/S/C), and the admissible record-screen transports **(a)** are **linear** maps preserving the screen norm — so the

Banach–Lamperti classification applies — and **(b)** act **transitively** on the screen sphere: any normalized configuration is reachable from any other (any-to-any reachability, strictly stronger than mere reversibility/invertibility of a transport). Under (a) and (b), Banach–Lamperti forces $q = 2$ — hence Born’s square-root composition — by a named classical theorem. (In finite dimension every injective linear isometry is automatically onto, so the surjectivity hypothesis of Banach–Lamperti is free; the equal-split argument is taken at screen size $n \geq 2$, the $n = 1$ sphere being degenerate.) Linearity is load-bearing, not cosmetic: drop it and *every* ℓ_q sphere becomes a homogeneous space under (nonlinear) transitive transports — an explicit $q = 3$ transport carries $(1, 0, 0)$ to the uniform vector while staying on the sphere — so transitivity alone would not single out $q = 2$. This is the precise, minimal form of the import that the correspondence theorems otherwise carry wholesale.

Statement. For record ledgers in the tame class — sketched here since it is load-bearing: inductive towers of finite record processes whose generator matrix elements grow at most polynomially in the tower level, whose limiting quadratic forms are closable, and whose boundary behavior is limit-point in the Weyl sense (no boundary data needed); the full definition with the growth constants is corpus-bound [P], as is the boundary classification with its receipts [P] — the weight calculus reconstructs: (i) Born composition — amplitudes attach to seams as square roots of record weights, with relative seam data supplying interference (the silent/physical criterion for such data is stated in Part II, Section 2, where it carries superselection weight); (ii) the Weyl/CCR algebra over the reconstructed configuration observables, landing in the Schrödinger representation by Stone–von Neumann (named import); (iii) unbounded position/momentum with ladder structure and selection rules.

Receipts. Gaussian record towers reconstruct continuum spectra from finite data with errors falling from 7×10^{-2} to 4.5×10^{-11} along the tower; dilation towers are exact and projectively coherent; the boundary of the tame class is *itself classified* — non-tame towers are exactly those missing boundary records, realizing the classical Weyl limit-point/limit-circle alternative [16] on record towers, with the threshold case appearing at an explicit $1/\log n$ merging rate.

Scope. The reconstruction half of the program’s “L3” problem is closed *conditionally on the [P] tame-class definition* (whose growth constants are corpus-bound); beyond-tame self-adjointness is classified, not derived. (The frontier taxonomy of Section 7 tracks this.)

3. Lorentz signature and the operational metric

Statement, with its honest scope. The commitment structure of records — the seal order is unrefinable while relational directions are not — *is argued to select a Lorentzian split*: exactly one timelike direction, with orientation classes that follow from the same commit-order argument (on the same [P] footing, not independently proved). This is a structural argument from the commit-order asymmetry (the seal order contributes the one unrefinable direction, the relational directions the rest), not a theorem proved in this batch; the proof obligation — that an unrefinable seal order forces exactly one timelike direction in the reconstructed operational metric — is named here and is corpus-bound [P], on the same footing as the open Lorentzian-port obligation of §4. What it does **not** force, from the axioms alone, is the number of spatial dimensions. The honest status of $d = 3$, stated once and relied on everywhere: Part II proves that exchange holonomy trivializes for $d \geq 3$ and that $d = 2$ is the anyonic exception; *nothing in this batch excludes $d \geq 4$* . The selection is therefore a **minimality principle** — $d = 3$ is

the first dimension with trivial exchange holonomy — named here as an input, exactly parallel to the fiber-minimality input of Part III, and not silently absorbed; the audit table types it accordingly. We note the instructive contrast with causal set theory [12], where dimension is likewise emergent-or-measured rather than axiomatic. The emergent metric is operational: the lapse function *is* the local clock rate of record formation.

Receipts. The lapse-equals-clock-rate identification is internally consistent in curved record geometries to ratio error 3.7×10^{-8} ; 1+1-dimensional Friedmann–Robertson–Walker redshift to 2.1×10^{-4} ; a first genuinely 3+1 curved instance (32^3 spatial lattice) whose agreement is at this stage qualitative (phase-quadrature structure reproduced; the quantified receipt is corpus-bound [P] and we quote no number for it here). The Euclidean–Lorentzian bridge is *operational rather than axiomatic*: the Osterwalder–Schrader rotation is implemented as a theorem-backed construction (record kernels reconstruct generators to 3.6×10^{-11} ; Lorentzian leapfrog dynamics matches OS propagators at the exact discretization order).

The continuum caveat, stated up front. What emerges is a *synthetic* geometry whose smooth Lorentzian manifolds form a regularity stratum. The program’s continuum theorems are class-uniform convergence results (a Mosco-convergence theorem with audited constants and a sharp rate [P]) plus a *detector characterization* of the smooth stratum (the graded local Weyl law with constructive converse of [C4] — proved there in one dimension with an executed two-dimensional port; the Lorentzian port is a named sequel, so the detector characterization of an emergent 3+1 Lorentzian geometry is a stated target, not a possessed theorem); homogenized record microstructure is *classified* as its own stratum, not mistaken for smoothness.

4. The arrow of time: reflection positivity as a theorem

Statement. For eventless sectors (no committed events in the relevant window) admitting a finite primitive Markov presentation, reflection positivity *holds as a theorem* — the ledger cannot hide a thermodynamic arrow. Quantitatively, the entropy-production rate is exactly the order-evidence divergence:

$$\sigma = \mathbb{E}[A_D] = D(P_{AB} \parallel P_{BA}),$$

the Kullback–Leibler divergence between forward and reversed pair statistics — “the arrow is the evidence.” This identity is *classical stochastic thermodynamics*, not a discovery of this program: entropy production as the relative entropy of forward versus time-reversed path measures goes back to Schnakenberg and is developed by Gaspard, Kawai–Parrondo–Van den Broeck, and Roldán–Parrondo [15]; we import it (the audit row types it as a classical import) and add only the record-pair form and the corollary that matters here — a **no-silent-arrow theorem**: eventless dynamics forces detailed balance, so a thermodynamic arrow cannot hide in a sector with no committed events.

Sharpness. The presentation hypothesis cannot be dropped: there exists an explicit three-dimensional process — the **record clock** — that is reversible, fully valid (all word probabilities nonnegative, certified analytically to all orders), of Hankel rank exactly three, whose diagonal oscillates irrationally on its decay circle. By Perron–Frobenius obstruction theory it admits *no* finite positive realization, and it *fails* reflection positivity: the reflected Gram on the 15-word family — the $1 + 2 + 4 + 8 = 15$ words of length ≤ 3 over a binary alphabet — has

minimum eigenvalue -1.06×10^{-2} (verified in companion [C2], Prop. 3.2; the eigenvalue and the Perron–Frobenius non-realizability are not reproduced in this paper). Sealed ledgers are thus a strict subclass of valid reversible processes, and the separating witness is precisely a clock — an object that keeps time without recording it. The structural question this opens (does reflection positivity plus finite rank *force* realizability?) is developed in a dedicated companion paper, where the obstruction is confined to combinatorially chiral patterns and reduced to two sharp finite-dimensional conjectures.

5. Thermality: the consistency credential

The horizon sector — temperature $T = \kappa/2\pi$ obtained two ways (Euclidean cone regularity, with its record-language reading as a no-boundary-record condition graded in [C-VIII] as a named identification, not a derived consequence of axiom S; Bisognano–Wichmann modular data on the record lattice), endpoint classification of lapse profiles, scattering, and capacity — is presented in full, with its receipts and with the classical literature it reproduces explicitly credited (Gibbons–Hawking, Calabrese–Cardy, Weyl endpoint theory), in the companion paper [C-VIII]. We do not duplicate it here; its role for this paper is one sentence: the record framework reproduces the standard thermal structure of horizons in exact lattice constructions, which is a consistency credential every candidate framework must earn and nothing more.

6. The number: one fixed-point constant

Axiom C is not decorative: it has a computable fixed point. Two registers, kept honestly separate: the *mathematics* of the fixed point — the equation, the constants $\theta, \eta, \kappa_b, \varepsilon$, their closed forms and exact algebraic relations — is fully self-contained in the submitted companion [C0]. The *identification* of that fixed point with axiom C’s refinement construction (the derivation sketched next) remains corpus-bound [P]: [C0] derives the same equation from its own exponential-consistency condition, and the claim that axiom C’s diamond construction produces that condition is a [P] dependency, graded as such in the audit. The derivation, in outline: refine one record into a binary tree of sub-records and demand that the committed correlation across the refinement diamond reproduce the unrefined correlation — projective consistency under binary refinement. For the minimal (2×2) committed diamond this closes into the single equation

$$\theta^3 + \theta^2 + \theta = 1,$$

whose unique real root $\theta = 0.543689\dots$ also satisfies $\tanh \eta = e^{-\eta}$ under $\theta = e^{-\eta}$ and the block-spin identity $\tanh(\eta/2) = \theta^2$ — one constant, three structural faces, with the equivalences verified symbolically and numerically. A binding analysis on the relation-code energetics of records (the constant κ_b — written with the subscript in this paper only to keep κ free for the surface gravity of result (iv); it is exactly [C0]’s κ — is the single-relation first-order marginality coefficient: the coefficient in [C0]’s law $\theta^w(1 - w\kappa_b)$, an *anti-binding* first order whose $w = 3$ sign is overturned at second order, per [C0] — with closed form $\kappa_b = \eta(1 - \theta^2)/(1 - \theta^2 + \theta)$; full self-contained definition, computation, and exact algebraic relations in the submitted companion [C0]) yields the program’s small parameter $\varepsilon = 3\kappa_b - 1 = 0.0317686364466$ (carried to ≥ 10 significant digits, the precision relevant at next-generation experimental sensitivity). Two things are claimed and no more: the constant’s pedigree contains no Standard-Model

input — checkable *within this batch* via [C0] — and its provenance predates by many program papers its registered confrontations with data (the accompanying note’s undressed spectrum-point registration and the mechanism paper [IX]’s dressed menu; both with stated falsifiers and named resolving experiments). Whether that confrontation survives is an experimental question; this section’s content is the derivation chain and its timestamp.

7. The audit: what is assumed, what is proved

The program maintains a complete dependency audit. Abbreviated:

result	pedigree
Born layer at tame scope	theorem (R,S,C) + Barandes correspondence import [1,2] (uniqueness of the quadratic calculus is inherited, not derived here) + Stone-von Neumann import + tame premise [P] + power-sum weight ansatz INPUT (componentwise additivity, per- component phase-blindness, single exponent; not derived from R,S,C)
Lorentzian split, orientation	structural argument [P] for the (1,d) split; spatial $d = 3$ INPUT (minimality principle, Sec. 3; $d \geq 4$ not excluded anywhere in this batch)
RP for eventless sectors	theorem + Markov-presentation premise (SHARP: the clock [C2])
$\sigma = \text{KL identity}$	classical import [15] (stochastic thermodynamics); program adds the record-pair form + no-silent-arrow
thermality $T = \kappa/2\pi$	receipts at stated scope [C-VIII], reproducing Gibbons-Hawking/BW structure (credited there)
lapse = clock rate	identification (operational, model-internal check $3.7\text{e-}8$)
continuum smooth stratum	class-uniform theorems [P] + detector characterization [C4] (1d + 2d port; Lorentzian open)
the constant θ , ϵ	equation + constants + closed forms: companion [C0] (in-batch, checkable); the axiom-C diamond reading of that equation: [P]; + two identifications in the binding chain [P]

The four scope premises promised in Section 1, counted here so the numbers reconcile: the tame-class premise [P]; the Markov-presentation premise (sharp, witness in [C2]); the finite/atomic sector premise; and the smooth-stratum regularity premise [P]. The three named structural inputs, likewise counted: the power-sum weight form (§2 input (0)); screen-transport linearity and transitivity (§2 inputs (a),(b)); and the spatial $d = 3$ minimality selection (§3). Named identifications and imports are enumerated exhaustively in the program’s audit paper; nothing is

silently load-bearing. The frontier (tame class, continuous sector families) is likewise structured: results are typed as rank-free, engine-uniform, finite-receipted, or genuinely frontier-bound.

8. Relation to prior work

The stochastic–quantum correspondence is Barandes’ [1, 2]; this program is an ontology *proposal within* that correspondence — records replace trajectories, and the axioms do forcing work the correspondence alone does not attempt. Among competing discrete or stochastic ontologies: ’t Hooft’s cellular-automaton interpretation [13] posits deterministic substrate dynamics where we posit sealed stochastic records; Nelson’s stochastic mechanics [14] derives the Schrödinger equation from diffusion but inherits its known locality difficulties; causal set theory [12] shares the discrete, order-theoretic spirit and the lesson that dimension must be earned rather than assumed. Reflection positivity as the encoding of the arrow connects to the Osterwalder–Schrader tradition [3]; the sharpness witness (the clock) places the program’s realization question inside the classical positive-realization literature (Dharmadhikari, Heller, Benvenuti–Farina — engaged in detail in [C2], where the witness and its certificates are constructed). Modular thermality belongs to Bisognano–Wichmann [4] and algebraic QFT [5]; the synthetic-geometry stance to metric-measure Lorentzian frameworks [6]. What we believe is new: the *direction* of derivation (positivity and the Lorentzian split as outputs of a record principle), the sharpness witnesses, and the audit discipline.

9. Limitations and the program’s frontier

Stated plainly: (i) all results hold at *stated scope* — tame class, finite/atomic sector structure, with the frontier taxonomy published; (ii) the continuum is reached as a regularity stratum, not assumed — and the multidimensional rigidity theorem is open; (iii) interacting dynamics (coupling flows, mass generation) is the program’s open dynamics layer; (iv) the identifications — few and named — are bridges, not theorems. On falsifiability we keep two notions separate: a *failing receipt* shows that the construction does not realize an identification — an internal refutation — while *empirical* falsification requires data. The batch’s empirical exposure is carried by two documents: the registered **undressed spectrum point** of the accompanying note [VII], and the mechanism paper [IX], which registers the dressed **four-member menu** (two members live against today’s data, two excluded and listed as falsified) with its own pre-committed kill criteria. Both are named here so this paper’s audit and the batch’s actual empirical surface agree.

Reproducibility

Every numerical claim regenerates from fixed-seed scripts with bit-identical reruns; the receipts repository accompanies the program. This paper’s role is the precise statement of axioms, theorems, and scope; companion papers carry the full proofs and extended receipts for each layer.

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- [17] J. Lamperti, On the isometries of certain function-spaces, *Pacific J. Math.* **8** (1958) 459–466 (after S. Banach, *Théorie des opérations linéaires*, 1932) — for $1 < q < \infty$, $q \neq 2$ the surjective linear isometries of complex ℓ_q^n are the generalized (phase-)permutations; the finite-dimensional endpoint cases ℓ_1^n, ℓ_∞^n have the same (generalized-permutation) isometry group by the separate

classical classification (Banach 1932), not by Lamperti's L^p theorem; only $q = 2$ yields the transitive group $U(n)$.

Companion papers [C1]–[C4]: the mathematical companions submitted with this batch — [C1] the necklace-invariant paper; [C2] the reflection-positivity / realization paper (the clock witness); [C3] the moment-signature test; [C4] the graded local Weyl law.